I wrote my code myself. As such I will just document my testing here.

Addition:

Input	Expected Output	Output
101 5 10	106	106
10 111 2	1001	5
111 10 2	1001	5

Thought I had managed alternate bases, but apparently not.
I've now changed the way I was doing the carrying to actually function (I had forgotten to add the carry step),
and made it so the final value is in the original base.

Input	Expected Output	Output
101 5 10	106	106
10 111 2	1001	1001
111 10 2	1001	1001
243 1637 8	2102	2102
1 1 10	2	2
243 1523 6	2210	2210
0 0 2	0	0

Multiplication:

Input	Expected Output	Output
101 5 10	106 505	106 505
10 111 2	1001 1110	1001 1110
111 10 2	1001 1110	1001 1110
243 1637 8	2102 447075	2102 447075
1 1 10	2 1	2 1
243 1523 6	2210 512213	2210 512213
0 0 2	0 0	0 0

All of this is as expected.

Division:

Input	Expected Output	Output
101 5 10	106 505 20	106 505
10 111 2	1001 1110 0	1001 1110 0
111 10 2	1001 1110 11	1001 1110 11
243 1637 8	2102 447075 0	2102 447075 0
1637 243 8	2102 447075 5	2102 447075 5
1 1 10	2 1 1	2 1 1
243 1523 6	2210 512213 0	2210 512213 0
1523 243 6	2210 512213 4	2210 512213 4
0 0 2	0 0 0	0 0 0

All as expected. Returns 0 when dividing by 0.